day 13 | 250926 F

Let's integrate data.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv('../data/velocities.txt', sep='\t', names=['t', 'v'])
df
```

Out[2]:

	t	v
0	0	0.000000
1	1	0.069478
2	2	0.137694
3	3	0.204332
4	4	0.269083
...	...	...
96	96	0.223073
97	97	0.254244
98	98	0.283753
99	99	0.311479
100	100	0.337308

101 rows × 2 columns

In [45]:
```
def trap(x_data, y_data):
    # if type(x_data)==pd.core.series.Series:
    #     x_data = np.asarray(x_data)
    #     y_data = np.asarray(y_data)
    dx = (x_data[-1:]-x_data[0])/len(x_data)
    s = dx/2*(y_data[0]+y_data[-1])
    s = s + np.sum(y_data[1:-2]*dx)
    return(s)

def simp(func, a, b, steps):
    if steps%2!=0:
        steps = steps + 1
        print('i added one step')
    dx = (b-a)/steps
    x = np.linspace(a, b, steps+1)
```

```
    y = func(x)
    s = dx/3*(func(a)+func(b))
    s = s + dx/3*(4*np.sum(y[1:steps:2]) + 2*np.sum(y[2:steps-1:2]))
    return(s)
```

In [46]:
```
x = np.linspace(0,2, 10000)
y = x**4 - 2*x + 1
```

In [47]:
```
trap(x, y)
```

Out[47]: array([4.39696131])

In [27]:
```
trap(df['t'], df['v'])
```

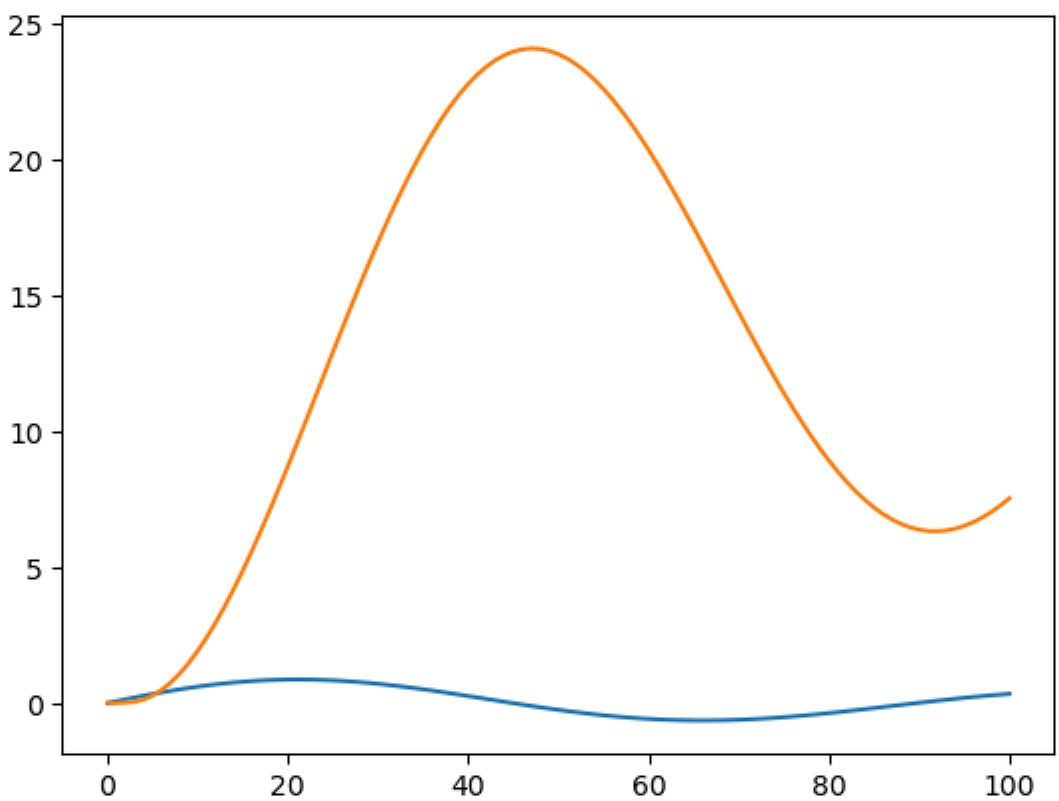
Out[27]: 7.8291534653465344

In [38]:
```
integrals = [0]


for i in range(1,101):
    w = trap(df['t'][0:i], df['v'][0:i])
    integrals.append(w)
```

In [39]:
```
fig, ax = plt.subplots()
ax.plot(df['t'], df['v'])
ax.plot(df['t'], integrals)
```

Out[39]: [<matplotlib.lines.Line2D at 0x76b1d4848050>]



In [48]:
```
integrals[-1]
```
Out[48]:
```
7.534669724999998
```
In [ ]:
```

```
In [44]:
```
df['t'][0]
```
Out[44]:
```
0
```
In [29]:
```
df['t'].index
```
Out[29]:
```
RangeIndex(start=0, stop=101, step=1)
```
In [33]:
```
range(1,10)
```
Out[33]:
```
range(1, 10)
```
In [ ]:
```

```